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 Studierichting: Informatica
 Oefeningenreeks 3D nr. 7.2

$$f(x) = e^x \cdot \sin x$$

$$*f'(x) = \frac{d}{dx} [e^x] \cdot \sin x + e^x \frac{d}{dx} [\sin x]$$

$$= e^x \cdot \sin x + e^x \cdot \cos x$$

$$= e^x (\sin x + \cos x)$$

$$*f'(x) = 0$$

$$\Rightarrow e^x (\sin x + \cos x) = 0$$

$$\Rightarrow \sin x + \cos x = 0$$

$$\Rightarrow \sin x = -\cos x$$

$$\Rightarrow \tan x = -1$$

$$\rightarrow x = -\frac{\pi}{4} + n\pi, n \in \mathbb{Z}$$

x		$-\frac{\pi}{4}$		$\frac{3\pi}{4}$		$\frac{7\pi}{4}$	
$f'(x)$	-	0	+	0	-	0	+
$f(x)$	$\searrow$	m	$\nearrow$	M	$\searrow$	m	$\nearrow$

References